

TraceMail AI: A Forensic Email Threat Investigation Platform with Multi-Source Threat Intelligence Correlation, Cryptographic Chain-of-Custody, and Explainable AI Scoring

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Abstract

Email remains the predominant attack vector for advanced persistent threats, credential harvesting, and Business Email Compromise (BEC). While traditional Secure Email Gateways (SEGs) provide binary perimeter filtering and Security Information and Event Management (SIEM) tools aggregate alerts, Security Operations Center (SOC) tier-1 and tier-2 analysts face a severe operational bottleneck: conducting deep-dive, per-message forensic investigations requires manually cross-referencing upwards of six disjoint intelligence portals, manually inspecting RFC 5322 headers, and reconstructing relay pathways.

This paper presents **TraceMail AI**, an end-to-end, web-based email forensic investigation and intelligence correlation platform. TraceMail AI automates raw `.eml` / RFC 822 file ingestion and live Google OAuth 2.0 mailbox streaming (`gmail.readonly`). The platform features: (1) a **Trusted-MTA Boundary** parsing engine that accurately identifies the originating public IP address while remaining resilient against attacker-forged `Received:` headers; (2) a **Canonical Threat Intelligence Gateway** that orchestrates parallel, fault-tolerant queries across VirusTotal v3, AbuseIPDB v2, IPinfo, URLScan.io, Google Safe Browsing v4, ICANN RDAP/WHOIS, and live DNS cryptographic resolvers (SPF, DKIM, DMARC); (3) a **Cryptographic Evidence Locker** generating SHA-256 integrity digests with active tamper detection for legal chain of custody; and (4) an **Explainable AI (XAI) Engine** combining mathematically grounded indicator contribution weights with Groq Cloud LLaMA-3 narrative synthesis.

The architecture is implemented using Next.js 15, FastAPI, and PostgreSQL 16 across Vercel and Railway cloud environments. We document the rigorous security hardening and audit phase that verified strict tenant isolation across all endpoints, resolved race conditions via row-level database locking, eliminated hardcoded simulation fallbacks in production, and validated the system against an 83-test automated regression suite. The system represents a fully functional prototype with live core investigation flows verified in production.

Keywords

Email Forensics, Phishing Detection, Business Email Compromise (BEC), Threat Intelligence Correlation, Header Analysis, Trusted MTA Boundary, Cryptographic Chain of Custody, Explainable AI, SPF/DKIM/DMARC Verification, Tenant Isolation.

1. Introduction

According to global cybersecurity threat reports, over 90% of organizational data breaches initiate via deceptive email communications. Modern threat actors rarely rely on generic mass-phishing templates; instead, they deploy sophisticated evasion techniques including display-name spoofing, lookalike/typosquatted domains, multi-hop open relay routing, clean-link redirect chains, and compromised vendor accounts.

When a suspicious email bypasses automated perimeter defenses and reaches a user's inbox, incident responders must perform forensic triage. In conventional SOC workflows, this procedure is heavily fragmented and labor-intensive:

1. The analyst downloads the raw `.eml` or extracts headers.
2. The analyst decodes RFC 5322 `Received:` headers top-to-bottom to isolate the sending Mail Transfer Agent (MTA), frequently misidentifying internal relays or attacker-forged headers as the true origin.
3. The analyst navigates between separate web services to verify DNS records (SPF TXT records, DKIM selectors, DMARC policies), query domain registration age via WHOIS/RDAP, check IP reputation on AbuseIPDB, submit embedded URLs to URLScan and Google Safe Browsing, and evaluate attachment hashes against VirusTotal.
4. The analyst synthesizes disparate threat scores into a cohesive incident report and establishes chain of custody for legal and compliance requirements.

This manual workflow requires significant analyst time per incident, creates cognitive fatigue, and introduces human error into threat triage.

Existing solutions exhibit critical limitations:

- **Secure Email Gateways (SEGs)** (e.g., Proofpoint, Mimecast, Microsoft Defender for Office 365) operate as inline filters that either deliver, quarantine, or drop an email. When an email slips past or is reported by an end-user, SEGs offer limited interactive forensic breakdown or open intelligence correlation to the analyst.
- **SIEM / SOAR Platforms** (e.g., Splunk, Microsoft Sentinel, Cortex XSOAR) ingest high-level alert metadata rather than the raw, interactive message structure, requiring costly enterprise licensing and custom playbook development that is beyond the budget of small and mid-sized enterprises (SMEs).

TraceMail AI was engineered to eliminate this forensic gap by providing an open, unified, explainable email investigation platform that automates extraction, intelligence aggregation, threat scoring, and forensic evidence sealing into a single responsive interface.

2. Related Work & Existing Systems

Capability / Dimension	Secure Email Gateways (SEGs)	SIEM / SOAR Platforms	Generic Intel Portals (e.g., VirusTotal GUI)	TraceMail AI Platform
Primary Focus	Inline block/allow perimeter filtering	Centralized log ingestion & alerting	Standalone indicator lookup	Deep-dive per-email forensic investigation
Header Spoofing Resilience	Proprietary internal rules	Dependent on parsed log fields	None (indicator-only)	Trusted-MTA Boundary Algorithm
Multi-Source Intel Aggregation	Vendor proprietary threat feeds	Requires complex custom playbooks	Single-source or manual multi-tab	Normalized 7-source live parallel gateway
Cryptographic Evidence Locker	Basic message archive	Log storage (tamper-evident audit)	None	SHA-256 RFC-822 hashing & tamper verification
Decision Explainability	Opaque spam/phishing score	Rule/query trigger matches	Raw engine detection counts	Mathematically grounded weights + LLaMA-3 narrative
Geospatial & Relay Visualization	Text table or absent	Custom dashboard plugins	Basic IP location	Full MTA hop timeline, GeoJSON map, Google Maps OSINT

While threat intelligence platforms like MISP and OpenCTI provide structured indicator storage, they lack direct parsing integration with live mailboxes and RFC-822 structures. TraceMail AI bridges the gap between raw email artifacts and standardized threat intelligence.

3. System Architecture

TraceMail AI employs a decoupled, cloud-native micro-architecture structured into five distinct operational tiers:



Tables (Railway)	7 Active APIs	`gmail.readonly`
+-----+	+-----+	+-----+

3.1 Presentation Tier

- **Framework:** Next.js 15 with Turbopack, React 19, TypeScript, and Tailwind CSS.
- **State Management:** Lightweight Zustand stores with synchronous `window.localStorage` token hydration (`authHeaders()`) ensuring immediate, non-racy authorization headers on all client-side requests.
- **Hosting:** Vercel edge deployment (`https://tracemail-ai-84ho.vercel.app`).
- **Core Views:**
 - `/dashboard` : EML file drag-and-drop upload, case registry, summary statistics.
 - `/inbox` : Live connected Gmail inbox feed, message scanner, on-demand investigation trigger.
 - `/investigation/[id]` : Comprehensive case analysis view featuring threat gauges, explainability breakdowns, authentication chips, raw header explorer, and Google Maps OSINT overlay.
 - `/soc` : SOC Command Center displaying threat metrics, attack distributions, and active monitoring alerts.
 - `/evidence` : Evidence Locker interface with SHA-256 seal verification, chain-of-custody event timeline, and one-click tamper simulation testing.
 - `/org` : Organizational threat heatmap visualizing department-level risk exposure.
 - `/reports` : Executive PDF and JSON report generation and export.

3.2 Application Tier (Backend API)

- **Framework:** FastAPI running on Python 3.12 and Uvicorn.
- **Deployment:** Railway containerized environment with health monitoring probes (`https://tracemail-ai-production.up.railway.app`).
- **API Surface:** 79 total registered routes across 13 modular API routers (`admin`, `ai_explainability`, `auth`, `campaigns`, `email`, `evidence`, `inbox`, `investigations`, `maps`, `org`, `reports`, `soc`, `threat`).
- **Dependency Injection:** Database sessions (`get_db`), strict authentication (`get_current_user`, `require_auth`), and canonical authorization boundaries (`verify_mailbox_access`).

3.3 Database Tier (Relational Schema)

The PostgreSQL 16 database consists of **16 normalized relational tables** enforcing data integrity and tenant boundaries:

1. `users` : System users with bcrypt-salted password hashing, email identifiers, and role-based permissions (`analyst`, `admin`).
2. `investigations` : Primary investigation records (39 columns) storing sender/recipient metadata, threat scores, verdicts, forensic markers, and foreign key `owner_user_id`.
3. `emails` : Raw RFC-822 message payloads and parsed body content.
4. `scans` : Background scan execution tracking and telemetry.
5. `ai_results` : Structured AI explainability reasoning and LLaMA-3 narrative summaries.
6. `headers` : Normalized email header key-value pairs.
7. `ioc_entities` : Extracted Indicators of Compromise (IPs, domains, hashes, URLs).
8. `threat_results` : Detailed output payloads from external threat intelligence providers.
9. `reports` : Generated forensic report metadata and export audit trails.
10. `audit_logs` : Administrative actions and security-relevant event logs.

11. `investigation_geo_cache` : 30-day cached geospatial coordinates and routing polylines.
12. `gmail_accounts` : OAuth 2.0 connected mailboxes, refresh tokens, and tenant ownership foreign key `owner_user_id` .
13. `inbox_scan_results` : Message-level scan outputs for connected mailboxes with unique constraint on `(account_email, message_id)` .
14. `evidence_records` : SHA-256 forensic hashes, legal custody logs, investigator tags, and tamper statuses.
15. `attachment_scans` : File hashes, MIME types, size metrics, and VirusTotal detection results.
16. `org_metrics` : Aggregated departmental risk statistics and threat counts.

3.4 Canonical Threat Intelligence Gateway

The platform interfaces with **seven live intelligence services** using asynchronous `httpx` workers wrapped in strict timeout boundaries (5–8 seconds per provider) with `asyncio.gather(return_exceptions=True)` to prevent cascading failures:

1. **VirusTotal v3**: Hashes of attachments and extracted URLs evaluated against 70+ commercial antivirus engines.
2. **AbuseIPDB v2**: IP abuse confidence scoring, total report counts, and distinct reporting categories.
3. **IPinfo / GeolIP**: Autonomous System Number (ASN), ISP identification, country, city, and geographical coordinates.
4. **URLScan.io**: Live URL sandbox analysis, HTTP redirection chains, effective URLs, and DOM metadata.
5. **Google Safe Browsing v4**: High-confidence threat lists covering malware, social engineering (phishing), and unwanted software.
6. **ICANN RDAP / WHOIS**: Authoritative domain registration data, registrar provenance, and domain age calculation.
7. **DNS Cryptographic Resolver**: Direct socket-level DNS queries via `dnspython` validating SPF TXT records, DKIM public keys, and DMARC enforcement policies (`p=reject` , `p=quarantine` , `p=none`).
8. **Groq Cloud (LLaMA-3)**: High-speed LLM inference engine generating executive threat narratives grounded strictly in the factual telemetry collected above.

4. Methodology & Forensic Pipeline

4.1 RFC 5322 Parsing & Identity Mismatch Detection

The parser extracts standard addressing headers (`From` , `To` , `Subject` , `Date` , `Reply-To` , `Return-Path` , `Message-ID`). It executes two automated checks for Business Email Compromise (BEC):

- **Display Name Spoofing Detection**: Evaluates the human-readable display name against a dictionary of known high-value brands (e.g., PayPal, Microsoft, Google, Amazon, State Bank of India). If the display name contains a protected brand string but the fully qualified domain name (FQDN) in the email address does not match authorized domain suffixes, the system flags brand impersonation.
- **Channel Mismatch Validation**: Compares `sender_domain` against `reply_to_domain` and `return_path_domain` . Inconsistencies commonly indicate reply-diversion attacks.

4.2 Trusted-MTA Boundary Origin-IP Extraction

The RFC 5321 / RFC 5322 email transmission standard dictates that every intermediary Mail Transfer Agent (MTA) prepends a `Received:` header to the top of the existing header block. In an unmanipulated message, the earliest hop appears at the bottom, and the final receiving hop appears at the top.

The Vulnerability: An attacker transmitting a phishing message can pre-inject arbitrary, fraudulent `Received:` headers before sending the email. For example, an attacker routing through `attacker.relay.net` (`198.51.100.55`) can append:

```
Received: from forged-victim.bank.com (1.2.3.4) by attacker.relay.net; Sun, 13 Sep 2026 11:59:59 +0000
```

Naive forensic parsers that inspect the bottom-most `Received:` line incorrectly extract `1.2.3.4` (an innocent financial institution) as the origin IP, completely exonerating the attacker's actual IP address (`198.51.100.55`).

The TraceMail AI Solution:

1. The parser maintains a compiled list of `TRUSTED_MTA_PATTERNS` corresponding to major receiving MX providers (`mx.google.com`, `gmail-smtp-in.l.google.com`, `mx.microsoft.com`, `protection.outlook.com`, `pphosted.com`, `mimecast.com`, `sendgrid.net`, `amazonses.com`).
2. The algorithm searches top-down to identify the **earliest verified trusted receiving boundary** (the lowest hop stamped by an authenticated destination MX server).
3. It extracts the client IP address recorded in the `from ... [IP]` clause of *that specific trusted hop*. Because this IP was stamped by the recipient's own gateway based on the direct TCP socket connection, the attacker cannot forge or tamper with it.
4. The candidate IP is validated against an RFC 1918 filter (`_is_public_ip`) to discard internal/private ranges (`10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`, `127.0.0.0/8`, multicast).
5. If no trusted boundary pattern matches (e.g., custom self-hosted mail servers), the engine traverses top-down from the recipient downwards, ensuring the final external hop is selected over attacker-controlled bottom entries.

4.3 Cryptographic DNS Authentication

Using `dnspython`, the platform verifies domain security records against authoritative nameservers:

- **SPF (Sender Policy Framework, RFC 7208):** Evaluates published TXT records against the extracted originating IP address.
- **DKIM (DomainKeys Identified Mail, RFC 6376):** Inspects header cryptographic signatures and public key selectors.
- **DMARC (RFC 7489):** Evaluates identifier alignment between the `From` domain and SPF/DKIM verification results, noting policy enforcement (`p=none`, `p=quarantine`, `p=reject`).

4.4 Mathematically Grounded Explainable AI (XAI) Scoring

Rather than relying on an ungrounded black-box classification, TraceMail AI calculates a deterministic **Phishing Threat Score (0–100)** through four weighted contribution categories:

- **Category A: Authentication Integrity (Weight: 30%)**
 - Cryptographic SPF Pass: `-10` to `+5` (safe baseline)
 - SPF Hard Failure / Softfail: `+15` to `+25`

- DKIM Failure / Missing: +15
- DMARC Alignment Failure: +20
- **Category B: Domain Age & Infrastructure (Weight: 20%)**
 - Newly Registered Domain (<30 days old): +25
 - Recent Domain (30–90 days old): +15
 - Established Domain (>365 days): 0
 - Display Name Brand Spoofing Mismatch: +30
- **Category C: Threat Intelligence Feeds (Weight: 30%)**
 - VirusTotal Malicious URL/Attachment Detections: +10 per engine (capped at +35)
 - AbuseIPDB Confidence of Abuse Score (>50%): +15 to +30
 - Google Safe Browsing Threat Match: +35
 - URLScan Malicious Verdict: +25
- **Category D: Content & Structural Heuristics (Weight: 20%)**
 - Urgent/Coercive Call to Action Language: +10
 - Credential Harvesting Keyword Density: +15
 - High-Risk Executable Attachments (.exe , .scr , .vbs , .iso): +25

The final score categorizes the email into:

- **Safe:** 0 – 34
- **Suspicious:** 35 – 64
- **Phishing / Critical Threat:** 65 – 100

Each scored factor is emitted as a discrete reason object containing category, weight, label, and description. This structured telemetry is subsequently fed to the **Groq LLaMA-3 AI Engine** with a strict system prompt instructing it to synthesize an executive incident summary *strictly grounded in the computed evidence*, eliminating generative hallucinations.

5. Security & Data Integrity Considerations

During the platform's multi-phase security hardening and audit cycle, critical architectural improvements were implemented:

5.1 Strict Multi-Tenant Isolation & Authorization Boundaries

In multi-user cybersecurity environments, unauthorized cross-tenant data access is a catastrophic failure mode. The audit identified and addressed the following security requirements:

- **Canonical Ownership Helpers:** In early iterations, access control was inconsistently applied across list routes versus detail lookups. This was replaced by a universal canonical helper: `verify_mailbox_access(account_email, current_user, db)` and `_verify_investigation_access(inv, current_user)`.
- **Default-Deny Enforcement:** Non-admin analysts can only access investigations and mailboxes where `owner_user_id == current_user.id` or where the connected mailbox matches their verified normalized

email. Any attempt by a user to query or scan another tenant's mailbox immediately yields a `403 Forbidden` response.

- **Route Protection Audit:** Every administrative route (`/api/v1/admin/analytics` , `/api/v1/admin/audit-logs`) strictly enforces `current_user.role == 'admin'` (returning 401 for unauthenticated requests and 403 for non-admin analysts). All maps, explainability, and evidence routes are authenticated with zero public leaks.

5.2 Elimination of Mock Fallbacks in Production

A critical milestone of the audit was purging legacy demonstration bypasses. Under production operation:

- `ENVIRONMENT=production`
- `USE MOCK THREAT INTEL=false`
- `ENABLE_DEMO_SEED=false` Live accounts never silently fall back to mock data corpora. If an external threat intelligence provider experiences an outage, the system explicitly returns structured error indicators rather than synthesizing simulated positive detections.

5.3 Concurrency & Idempotency Controls

Under high-volume SOC triage, multiple analysts or automated processes might trigger simultaneous scans on the same inbox message. To eliminate race conditions and database constraint violations (`uq_inbox_scan_account_message`), the investigation service utilizes PostgreSQL row-level locks:

```
db.query(Investigation).filter(Investigation.id == id).with_for_update().first()
```

This guarantees that concurrent scan requests execute idempotently without corrupting forensic records or duplicating investigation cases.

5.4 Cryptographic Chain of Custody & Evidence Sealing

In `EvidenceService` , upon initial RFC-822 ingestion, a cryptographic SHA-256 hash is computed across the raw headers and body payload: `$$\text{Evidence Hash} = \text{SHA-256}(\text{Raw Headers} \parallel \text{Body Content})$$` This value is written to the immutable `evidence_records` table alongside investigator credentials, ISO-8601 timestamps, and initial custody notes. The platform exposes an automated verification endpoint (`POST /api/evidence/{id}/verify`) that re-hashes the stored record against the original seal, enabling automated detection of bit-level data tampering or database corruption.

6. Implementation & Verification Testing

The backend system is validated by an automated test suite comprising **83 unit and integration tests** executing under `pytest` .

```
===== 83 passed in 39.16s =====
```

6.1 Test Suite Breakdown

1. `test_email.py` : RFC-822 parsing accuracy, legitimate email benchmark validation (Internshala sample), and BEC display-name spoofing detection.
2. `test_inbox_investigate.py` : Root Cause 8 verification (Trusted-MTA boundary extraction preventing IP forging), concurrency testing, and demo fallback removal verification.

3. `test_reports.py` : Detail schema validation, GeoJSON map output contracts, timeline step generation, and PDF/JSON export generation.
4. `test_threat_apis.py` : Threat intelligence client timeout handling, provider normalization schemas, and graceful degradation under API failures.
5. `test_v2_features.py` : Google OAuth status checks, inbox scan/results endpoints, SOC metrics aggregation, Evidence Locker SHA-256 seal verification and simulated corruption detection, AI explainability weight mathematical sums, and attachment malware detection.
6. `test_maps_services.py` : IP geolocation caching (30-day cache validation), Google Maps polyline decoding, Places infrastructure OSINT querying, and authenticated route protection.
7. `test_campaigns.py` : Campaign threat clustering and detail view access controls.
8. `test_narrative_consistency.py` : Verification that Groq LLaMA-3 narrative summaries agree with deterministic DNS/reputation indicators.

6.2 Frontend Production Verification

The frontend TypeScript codebase was validated via Next.js production build compilation (`next build`), confirming zero type errors, strict interface compliance, and proper bundle tree-shaking.

7. Results & Current Deployment Status

The TraceMail AI platform is deployed and fully operational in a production environment:

Component / Service	Target Environment	Live URL / Endpoint	Verification Status
Web User Interface	Vercel Edge	<code>https://tracemail-ai-84ho.vercel.app</code>	Operational (Verified)
Backend REST API	Railway Cloud	<code>https://tracemail-ai-production.up.railway.app</code>	Operational (Verified)
PostgreSQL Database	Railway Cloud	<code>altaria.proxy.rlwy.net:16589</code>	16 Tables Live (Verified)
Threat Intel Gateway	Multi-Provider	<code>/health/apis</code>	9/9 Providers Live (Verified)
VirusTotal v3	Commercial API	Integrated via <code>vt_client</code>	Live (Verified)
AbuseIPDB v2	Commercial API	Integrated via <code>abuse_client</code>	Live (Verified)
IPinfo Geolocation	Commercial API	Integrated via <code>geo_client</code>	Live (Verified)
URLScan.io	Sandbox API	Integrated via <code>urlscan_client</code>	Live (Verified)
Google Safe Browsing	Google Cloud API	Integrated via <code>gsb_client</code>	Live (Verified)
Groq LLaMA-3	Cloud AI API	Integrated via <code>groq_client</code>	Live (Verified)
Google Maps Platform	Google Cloud API	Integrated via <code>maps_service</code>	Live (Verified)
Gmail OAuth2	Google Cloud Console	Ingestion via <code>inbox_service</code>	Live (Verified)
Tenant Isolation	Database / App	All routes checked (0 admins)	Enforced (Verified)
Evidence Locker	Cryptographic DB	SHA-256 + Tamper Verification	Enforced (Verified)

All 32 primary API routes enforce appropriate authentication; all mailbox and investigation endpoints utilize canonical ownership authorization helpers; database integrity audits confirm zero orphaned (`NULL owner_user_id`) records; and environment configurations report `ENVIRONMENT=production`, `USE MOCK THREAT INTEL=false`, and `ENABLE DEMO SEED=false`.

8. Limitations & Future Work

To maintain rigorous academic and engineering honesty, the following constraints and areas for future development are acknowledged:

- Performance Benchmarking:** While individual external API timeouts are strictly enforced (5–8s), formal end-to-end throughput and latency benchmarks (e.g., maximum concurrent investigations per second under full load) have **not yet been benchmarked** with standardized load-testing tools (such as Locust or JMeter).
- Rate-Limiting Economics:** The threat intelligence gateway currently depends on free/developer-tier API quotas. In an enterprise setting with thousands of emails per hour, a distributed caching layer (Redis) and commercial quota pooling are required to prevent provider throttling.

3. **Role-Based Access Control (RBAC):** The current model enforces basic binary roles (`admin` vs. `analyst`). Future iterations will implement fine-grained multi-tenancy supporting organization hierarchies, team leads, read-only auditors, and client tenant silos.
 4. **Distributed Worker Architecture:** Deep forensic parsing currently executes within asynchronous FastAPI event loops. Future releases will offload heavy file analysis and sandboxing to distributed task queues (e.g., Celery, Redis Queue, or Apache Kafka).
 5. **Deployment Topology:** The platform is currently hosted in single-region container infrastructure. Multi-region redundancy and high-availability database clustering represent future operational objectives.
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9. Conclusion

TraceMail AI demonstrates that manual, error-prone email threat investigations can be successfully transformed into an automated, explainable, and cryptographically verified forensic workflow. By combining a spoofing-resistant Trusted-MTA boundary extraction algorithm with parallel multi-source threat intelligence, mathematically grounded explainability scoring, and SHA-256 evidence sealing, the platform provides security analysts with rapid, transparent, and legally defensible email forensics. The platform's open micro-architecture, validated across 83 automated tests and live cloud deployments, establishes a viable foundation for modern SOC operations and accessible cybersecurity defense.

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